

# Homework 3 - ET4369 Nyquist Rate Data Converters

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# 1 Spectral Performance Metrics

**Given:** ADC output data (`hw3_adc_out.mat`) with  $f_s = 100$  MS/s. The input was a full-scale sinusoidal signal. The ADC has a 10-bit bipolar mid-rise quantizer with full-scale range  $[-1, +1]$ .

## 1.1 Part (a): Spectrum Plot

The one-sided amplitude spectrum is computed from the 10000-point FFT. Amplitudes are normalized to dBFS using the full-scale amplitude  $A_{FS} = 1$  V:

$$X_{\text{dBFS}}(k) = 20 \log_{10} \left( \frac{|X(k)| \cdot 2/N}{A_{FS}} \right) \quad (1)$$

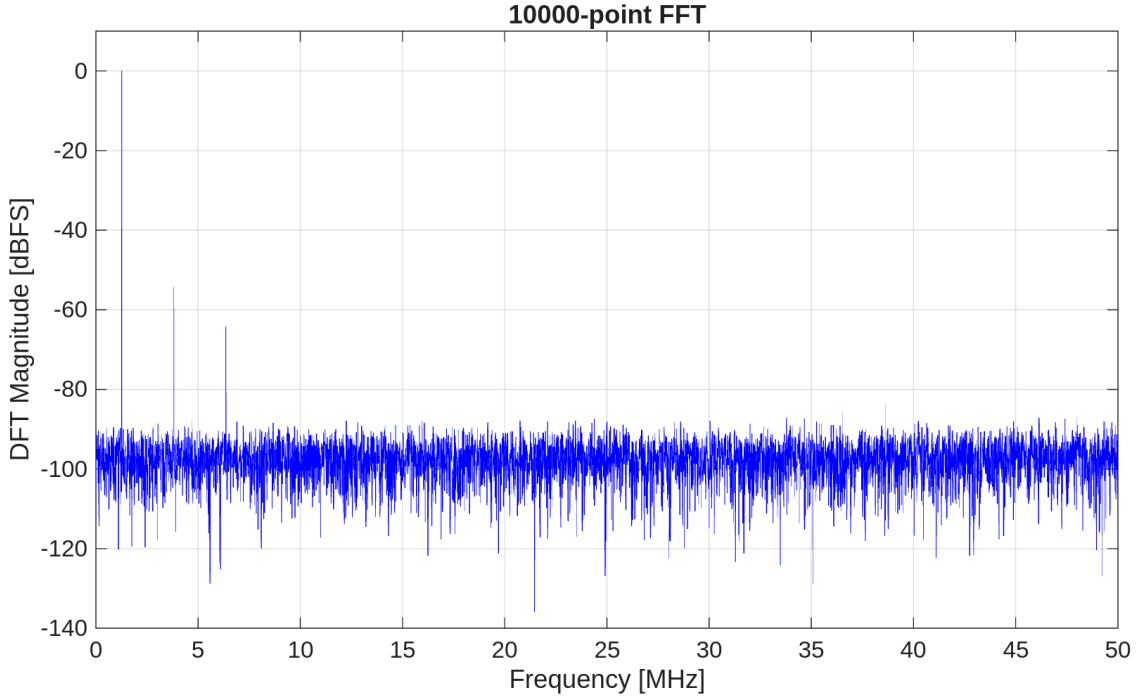


Figure 1: Output spectrum of the 10-bit ADC (10000-point FFT), showing the signal tone at 1.27 MHz, HD3 at 3.81 MHz, and HD5 at 6.35 MHz.

## 1.2 Part (b): Input Frequency

The signal frequency is identified from the FFT bin with the largest magnitude (excluding DC):

$$f_{\text{in}} = 1.27 \text{ MHz} \quad (\text{bin } 127) \quad (2)$$

The signal level is approximately 0 dBFS, confirming a full-scale input.

## 1.3 Part (c): Performance Metrics

Metrics are computed using standard IEEE definitions. Harmonics 2 through 7 are identified and their power is separated from noise.

Table 1: Harmonic levels in the ADC output spectrum.

| Harmonic | Frequency (MHz) | Level (dBFS) | Notes                |
|----------|-----------------|--------------|----------------------|
| HD2      | 2.54            | −106.5       | Even-order           |
| HD3      | 3.81            | −54.5        | <b>Dominant spur</b> |
| HD4      | 5.08            | −90.4        | Even-order           |
| HD5      | 6.35            | −64.3        | 2nd largest spur     |
| HD6      | 7.62            | −106.5       | Even-order           |
| HD7      | 8.89            | −103.8       | —                    |

Table 2: Summary of spectral performance metrics.

| Metric | Value     |
|--------|-----------|
| SNR    | 58.94 dB  |
| SNDR   | 52.90 dB  |
| ENOB   | 8.50 bits |
| THD    | −54.15 dB |
| SFDR   | 54.58 dB  |

The ENOB of 8.50 bits (vs. 10-bit quantizer resolution) indicates significant degradation due to distortion and excess noise.

#### 1.4 Part (d): SFDR-Limiting Non-Ideality

The SFDR is determined by **HD3** (3rd harmonic distortion) at −54.5 dBFS. The dominance of odd-order harmonics (HD3 and HD5) is characteristic of a **symmetric (odd) nonlinearity**, which is typically caused by the signal-dependent on-resistance of the sampling switch:

$$r_{\text{on}}(V_{\text{in}}) = \frac{1}{\mu C_{\text{ox}} \frac{W}{L} (V_{\text{GS}} - V_t - V_{\text{in}})} \quad (3)$$

This produces a predominantly cubic transfer function error  $\propto V_{\text{in}}^3$ , generating HD3 and HD5.

#### 1.5 Part (e): Additional Noise Sources

Comparing measured noise to ideal quantization noise:

$$\text{SQNR}_{\text{ideal}} = 6.02 \times 10 + 1.76 = 61.96 \text{ dB} \quad (4)$$

$$\text{SNR}_{\text{measured}} = 58.94 \text{ dB} \quad (5)$$

The measured SNR is 3.0 dB lower than the ideal SQNR. Equivalently, the measured total noise power is approximately  $4.1 \times$  the ideal quantization noise power:

$$\frac{P_{\text{noise, total}}}{P_{\text{quant}}} = 10^{(61.96 - 58.94)/10} \approx 4.1 \quad (6)$$

**Yes, there are additional noise sources** beyond quantization noise. The electronic noise (thermal noise from switches, comparator noise, amplifier noise) contributes approximately  $3 \times$  the power of the quantization noise alone. The noise floor is elevated by about 3 dB compared to the ideal −98.95 dBFS floor.

## 2 Clock Bootstrapping

**Given:** Bootstrapped sampling switch with  $C_{\text{boot}} = C_L = 500 \text{ fF}$ ,  $C_{\text{par}} = 100 \text{ fF}$ ,  $V_{\text{DD}} = 1.8 \text{ V}$ ,  $V_t = 0.5 \text{ V}$ ,  $WLC_{\text{ox}} = 50 \text{ fF}$ ,  $\mu C_{\text{ox}} W/L = 10 \text{ mA/V}^2$ . Input range: 0.5 V to 1 V.

## 2.1 Circuit Operation

During  $\phi_1$  (precharge):  $C_{\text{boot}}$  charges to  $V_{\text{DD}}$ , gate of  $M_1$  is pulled to ground (off).

During  $\phi_2$  (track):  $C_{\text{boot}}$  bottom plate connects to  $V_{\text{in}}$  (source of  $M_1$ ), top plate connects to gate. By charge conservation at the gate node:

$$C_{\text{boot}}(V_G - V_{\text{in}}) + C_{\text{par}} \cdot V_G = C_{\text{boot}} \cdot V_{\text{DD}} \quad (7)$$

Solving for  $V_{\text{GS}}$ :

$$V_{\text{GS}} = V_G - V_{\text{in}} = \frac{C_{\text{boot}} \cdot V_{\text{DD}} - C_{\text{par}} \cdot V_{\text{in}}}{C_{\text{boot}} + C_{\text{par}}} \quad (8)$$

Without  $C_{\text{par}}$ , we would have  $V_{\text{GS}} = V_{\text{DD}}$  (perfectly bootstrapped). The  $C_{\text{par}} \cdot V_{\text{in}}$  term introduces a residual signal dependence.

## 2.2 Part (a): On-Resistance Variation

Using Eq. (8) and  $r_{\text{on}} = 1/(\mu C_{\text{ox}} W/L \cdot V_{\text{OV}})$  where  $V_{\text{OV}} = V_{\text{GS}} - V_t$ :

| $V_{\text{in}}$ (V) | $V_{\text{GS}}$ (V) | $V_{\text{OV}}$ (V) | $r_{\text{on}}$ ( $\Omega$ ) |
|---------------------|---------------------|---------------------|------------------------------|
| 0.5                 | 1.417               | 0.917               | 109.1 (minimum)              |
| 1.0                 | 1.333               | 0.833               | 120.0 (maximum)              |

$$\text{Percent increase} = \frac{r_{\text{on}}(1 \text{ V}) - r_{\text{on}}(0.5 \text{ V})}{r_{\text{on}}(0.5 \text{ V})} = \frac{120.0 - 109.1}{109.1} = 10.0\% \quad (9)$$

This can also be derived analytically. Since  $r_{\text{on}} \propto 1/V_{\text{OV}}$ :

$$\frac{r_{\text{on,max}}}{r_{\text{on,min}}} = \frac{V_{\text{OV}}(0.5 \text{ V})}{V_{\text{OV}}(1.0 \text{ V})} = \frac{11/12}{5/6} = \frac{11}{10} = 1.10 \quad (10)$$

## 2.3 Part (b): Charge Injection Error Variation

For fast gating with 50/50 channel charge split, the charge injected onto  $C_L$  when  $M_1$  turns off:

$$\Delta V_{\text{out}} = -\frac{WLC_{\text{ox}} \cdot V_{\text{OV}}}{2 \cdot C_L} = -\frac{50 \text{ fF}}{2 \times 500 \text{ fF}} \cdot V_{\text{OV}} = -\frac{V_{\text{OV}}}{20} \quad (11)$$

Since  $V_{\text{OV}}$  depends on  $V_{\text{in}}$  through Eq. (8), the error is signal-dependent:

| $V_{\text{in}}$ (V) | $V_{\text{OV}}$ (V) | $\Delta V_{\text{out}}$ (mV) |
|---------------------|---------------------|------------------------------|
| 0.5                 | 0.917               | -45.83                       |
| 1.0                 | 0.833               | -41.67                       |

$$\Delta V_{\text{out,pp}} = |-45.83 - (-41.67)| = 4.17 \text{ mV} \quad (12)$$

The signal-dependent part of the error arises from  $C_{\text{par}}$ : if  $C_{\text{par}} = 0$ , then  $V_{\text{OV}}$  would be constant ( $V_{\text{DD}} - V_t$ ), and the charge injection would be purely a fixed offset (no signal-dependent variation).

## 3 Relationship between INL and SFDR

**Given:** Output spectrum of an 11-bit A/D converter for a full-scale sinusoidal input (65536-point FFT shown on page 2 of the assignment).

### 3.1 Part (a): Maximum INL (Cubic Bow)

From the spectrum, the dominant spur is HD3 at approximately  $-40$  dBFS, giving SFDR  $\approx 40$  dB.

For a cubic INL with endpoint correction,  $\text{INL}(u) = A_3(u^3 - u)$  where  $u \in [-1, 1]$  is the normalized code. The maximum occurs at  $u = \pm 1/\sqrt{3}$  with value  $|u^3 - u|_{\max} = 2/(3\sqrt{3})$ . The relationship between INL and HD3:

$$\text{HD3} = \frac{3\sqrt{3} \cdot \text{INL}_{\max}}{4 \cdot 2^B} \quad (13)$$

Rearranging:

$$\text{INL}_{\max} = \frac{4 \cdot 2^B}{3\sqrt{3} \cdot 10^{\text{SFDR}/20}} = \frac{4 \times 2048}{5.196 \times 100} \quad (14)$$

$$\boxed{\text{INL}_{\max} \approx 15.8 \text{ LSB}} \quad (15)$$

### 3.2 Part (b): Electronic Noise Standard Deviation

The ideal quantization noise floor for an 11-bit ADC with a 65536-point FFT:

$$\text{Floor}_{\text{ideal}} = -\text{SQNR} - 10 \log_{10}(N/2) = -67.98 - 45.15 = -113.13 \text{ dBFS} \quad (16)$$

From the spectrum, the measured noise floor is approximately  $-100$  dBFS. The total noise power per bin is  $10^{13.13/10} \approx 20.6 \times$  the ideal quantization noise, meaning the electronic noise power is about  $19.6 \times$  the quantization noise.

$$\sigma_{\text{electronic}} = \Delta \sqrt{\frac{P_{\text{elec}}}{P_{\text{quant}}} \cdot \frac{1}{12}} = \Delta \sqrt{\frac{19.6}{12}} \quad (17)$$

$$\boxed{\sigma_{\text{electronic}} \approx 1.28 \text{ LSB}} \quad (18)$$

### 3.3 Part (c): Effective Number of Bits

The SNDR is dominated by distortion (HD3 at  $-40$  dBFS relative to the signal):

$$\text{SNDR} = 10 \log_{10} \left( \frac{P_{\text{signal}}}{P_{\text{noise}} + P_{\text{distortion}}} \right) \approx 39.9 \text{ dB} \quad (19)$$

Since the distortion power ( $10^{-4} \cdot P_{\text{signal}}$ ) greatly exceeds the noise power ( $\sim 10^{-5.5} \cdot P_{\text{signal}}$ ),  $\text{SNDR} \approx \text{SFDR}$ .

$$\boxed{\text{ENOB} = \frac{\text{SNDR} - 1.76}{6.02} = \frac{39.9 - 1.76}{6.02} \approx 6.3 \text{ bits}} \quad (20)$$

### 3.4 Part (d): INL and SFDR in a Commercial Product

**Paper:** S.-C. Lee et al., “A 10-bit 205-MS/s 1.0-mm<sup>2</sup> 90-nm CMOS Pipeline ADC for Flat Panel Display Applications,” *IEEE J. Solid-State Circuits*, vol. 42, no. 12, pp. 2688–2695, Dec. 2007.

From Fig. 14(b), the INL shows a dominant cubic nonlinearity with a smooth average amplitude of approximately  $\pm 0.9$  LSB. Estimating the SFDR from this INL:

$$\text{HD3} = \frac{3\sqrt{3} \times 0.9}{4 \times 2^{10}} = \frac{4.677}{4096} = 0.001142 \quad (21)$$

$$\text{SFDR}_{\text{estimated}} = -20 \log_{10}(0.001142) \approx 58.8 \text{ dB} \quad (22)$$

The paper specifies a typical SFDR of approximately 57 dB at 70 MHz input frequency.

$$\boxed{\text{SFDR}_{\text{from INL}} \approx 58.8 \text{ dB} \quad \text{vs.} \quad \text{SFDR}_{\text{specified}} \approx 57 \text{ dB}} \quad (23)$$

The 1.8 dB agreement confirms that the cubic INL is the dominant source of spurious tones in this ADC.

## 4 Switch Nonidealities

**Given:** Switched-capacitor circuit with all NMOS switches:  $W = 10 \mu\text{m}$ ,  $L = 0.2 \mu\text{m}$ ,  $C_{\text{ox}} = 10 \text{ fF}/\mu\text{m}^2$ ,  $V_t = 0.4 \text{ V}$ ,  $C_{\text{ol}} = 0.1 \text{ fF}/\mu\text{m}$ .  $C_1 = 1 \text{ pF}$ ,  $C_2 = 2 \text{ pF}$ . Fast gating assumed; settling dynamics ignored.

### 4.1 Device Parameters

| Parameter                              | Value                                                   |
|----------------------------------------|---------------------------------------------------------|
| $WLC_{\text{ox}}$                      | $10 \times 0.2 \times 10 = 20 \text{ fF}$               |
| $C_{\text{ov}}$ (overlap per terminal) | $C_{\text{ol}} \times W = 0.1 \times 10 = 1 \text{ fF}$ |

### 4.2 Circuit Operation

During  $\phi_1$ : left  $\phi_1$  switch connects  $1 \text{ V}$  source to  $V_1$  (charges  $C_1$ ), right  $\phi_1$  switch connects  $V_2$  to its reference. The  $\phi_2$  switch is off.

At  $t = t_1$ :  $\phi_1$  turns off (both  $\phi_1$  switches open simultaneously).

During  $\phi_2$ :  $\phi_2$  switch connects  $V_1$  to  $V_2$  (charge sharing between  $C_1$  and  $C_2$ ).

At  $t = t_2$ :  $\phi_2$  turns off.

### 4.3 Part (e): Standard Deviation of $V_1$ at $t_1$ and $t_2$

At  $t_1$  ( $\phi_1$  opens):  $V_1$  was tracking the  $1 \text{ V}$  source through the left  $\phi_1$  switch. The sampled  $kT/C$  noise:

$$\sigma(V_1)|_{t_1} = \sqrt{\frac{kT}{C_1}} = \sqrt{\frac{4.14 \times 10^{-21}}{1 \times 10^{-12}}} \quad (24)$$

$$\boxed{\sigma(V_1)|_{t_1} = 64.4 \mu\text{V}_{\text{rms}}} \quad (25)$$

At  $t_2$  ( $\phi_2$  opens): During  $\phi_2$ ,  $V_1$  and  $V_2$  were connected through the switch with both  $C_1$  and  $C_2$  floating. The total charge  $Q = C_1 V_1 + C_2 V_2$  is conserved. The noise on  $V_1$  decomposes into:

- **Common-mode noise** (from  $\phi_1$  sampling): The total charge carries noise  $\sigma^2(Q) = kT(C_1 + C_2)$ , contributing  $\sigma^2(V_{\text{cm}}) = kT/(C_1 + C_2)$  to  $V_1$ .
- **Differential noise** (from  $\phi_2$  switch): By equipartition,  $\sigma^2(\delta V_1) = kT \cdot C_2/[C_1(C_1 + C_2)]$ .

These are independent, so:

$$\sigma^2(V_1)|_{t_2} = \frac{kT}{C_1 + C_2} + \frac{kT \cdot C_2}{C_1(C_1 + C_2)} = \frac{kT}{C_1} \quad (26)$$

$$\boxed{\sigma(V_1)|_{t_2} = \sqrt{\frac{kT}{C_1}} = 64.4 \mu\text{V}_{\text{rms}}} \quad (27)$$

The result is the **same** at both times. This is a fundamental property of switched-capacitor circuits: the  $kT/C$  noise on a node depends only on the capacitance at that node, regardless of the switching history. The noise “reset” during  $\phi_2$  (charge sharing with  $C_2$ ) exactly compensates for the additional noise injected by the  $\phi_2$  switch.

#### 4.4 Part (f): Charge Injection on $V_1$ at $t_1$

When the left  $\phi_1$  switch turns off, the channel charge splits 50/50 (fast gating):

$$V_{GS} = 1.8 \text{ V} - 1.0 \text{ V} = 0.8 \text{ V} \quad (28)$$

$$V_{OV} = V_{GS} - V_t = 0.8 - 0.4 = 0.4 \text{ V} \quad (29)$$

$$Q_{ch} = WLC_{ox} \cdot V_{OV} = 20 \text{ fF} \times 0.4 \text{ V} = 8 \text{ fC} \quad (30)$$

Half of the channel charge is injected onto  $C_1$ :

$$\Delta V_1 = -\frac{Q_{ch}}{2 \cdot C_1} = -\frac{8 \text{ fC}}{2 \times 1 \text{ pF}} = -4.0 \text{ mV} \quad (31)$$

$$\boxed{V_1(t_1) = 1.0 - 0.004 = 0.996 \text{ V}} \quad (32)$$

#### 4.5 Part (g): Clock Feedthrough on $V_1$ at $t_1$

When  $\phi_1$  drops from 1.8 V to 0 V, the gate voltage change couples to  $V_1$  through the overlap capacitance  $C_{ov}$ :

$$\Delta V_1 = \Delta V_{gate} \cdot \frac{C_{ov}}{C_{ov} + C_1} = (-1.8 \text{ V}) \cdot \frac{1 \text{ fF}}{1 \text{ fF} + 1000 \text{ fF}} \quad (33)$$

$$\Delta V_1 = -1.8 \times \frac{1}{1001} = -1.80 \text{ mV} \quad (34)$$

$$\boxed{V_1(t_1) = 1.0 - 0.00180 = 0.99820 \text{ V}} \quad (35)$$

Note that charge injection ( $-4.0 \text{ mV}$ ) is the larger effect compared to clock feedthrough ( $-1.8 \text{ mV}$ ). In practice, both contribute simultaneously, giving a total error of approximately  $-5.8 \text{ mV}$ .

## 5 Boxcar Sampling

**Paper:** H. Gao, R. M. Walker, P. Nuyujukian, K. A. A. Makinwa, K. V. Shenoy, B. Murmann and T. H. Meng, “HermesE: A 96-Channel Full Data Rate Direct Neural Interface in  $0.13 \mu\text{m}$  CMOS,” *IEEE J. Solid-State Circuits*, pp. 1043–1055, Apr. 2012.

### 5.1 Part (h): Derivation of Equation (2)

Starting from Eq. (1) of the paper, the boxcar integrator output is:

$$Q_{in2,CT}(t) = \int_{t-DT_s}^t I_{out1d}(t') dt' \quad (36)$$

where  $D$  is the duty cycle of the  $\Phi_1$  sampling phase and  $T_s = 1/f_s$  is the SC clock period.

This is a convolution of  $I_{out1d}(t)$  with a rectangular window:

$$h(\tau) = \begin{cases} 1, & 0 \leq \tau \leq DT_s \\ 0, & \text{otherwise} \end{cases} \quad (37)$$

Taking the Fourier transform:

$$H(\omega) = \int_0^{DT_s} e^{-j\omega\tau} d\tau = \frac{1 - e^{-j\omega DT_s}}{j\omega} \quad (38)$$

Factoring out  $e^{-j\omega DT_s/2}$ :

$$\begin{aligned} H(\omega) &= e^{-j\omega DT_s/2} \cdot \frac{e^{j\omega DT_s/2} - e^{-j\omega DT_s/2}}{j\omega} \\ &= e^{-j\omega DT_s/2} \cdot \frac{2 \sin(\omega DT_s/2)}{\omega} \end{aligned} \quad (39)$$

Since  $Q_{\text{in2,CT}}(\omega) = I_{\text{out1d}}(\omega) \cdot H(\omega)$ , taking the magnitude:

$$\boxed{\frac{|Q_{\text{in2,CT}}(\omega)|}{|I_{\text{out1d}}(\omega)|} = \frac{2 \left| \sin\left(\frac{\omega DT_s}{2}\right) \right|}{\omega}} \quad (40)$$

This is equation (2) of the paper. In the baseband ( $\omega DT_s/2 \ll 1$ ), the sinc response is approximately flat, providing inherent anti-alias filtering by attenuating high-frequency noise components before they fold into the signal band.

## 5.2 Part (i): Verification of the 3.4× ENBW Advantage

### 5.2.1 Boxcar (sinc) prefilter ENBW

The power transfer function is  $|H(f)|^2 = (DT_s)^2 \text{sinc}^2(f \cdot DT_s)$ , with DC gain  $|H(0)|^2 = (DT_s)^2$ . The effective noise bandwidth:

$$\text{ENBW}_{\text{boxcar}} = \frac{1}{|H(0)|^2} \int_0^\infty |H(f)|^2 df = \int_0^\infty \text{sinc}^2(f \cdot DT_s) df = \frac{1}{2 DT_s} \quad (41)$$

using the identity  $\int_0^\infty \text{sinc}^2(u) du = 1/2$ .

### 5.2.2 Standard RC voltage sampling ENBW

For a conventional sample-and-hold with 0.1% dynamic settling accuracy:

$$e^{-N} = 0.001 \quad \Rightarrow \quad N = \ln(1000) = 6.908 \text{ time constants} \quad (42)$$

With the same available acquisition time  $DT_s$ , the time constant is  $\tau = DT_s/N$ , giving:

$$\text{ENBW}_{\text{RC}} = \frac{1}{4\tau} = \frac{N}{4 DT_s} \quad (43)$$

### 5.2.3 Ratio

Dividing Eq. (43) by Eq. (41):

$$\frac{\text{ENBW}_{\text{RC}}}{\text{ENBW}_{\text{boxcar}}} = \frac{N/(4 DT_s)}{1/(2 DT_s)} = \frac{N}{2} = \frac{6.908}{2} \quad (44)$$

$$\boxed{\frac{\text{ENBW}_{\text{RC}}}{\text{ENBW}_{\text{boxcar}}} = 3.45 \approx 3.4 \times} \quad (45)$$

The boxcar prefilter achieves 3.4× lower effective noise bandwidth compared to a conventional RC-based voltage sampling circuit, for the same acquisition time and 0.1% settling accuracy. This translates directly to a 3.4× reduction in aliased noise power, or equivalently a  $10 \log_{10}(3.4) = 5.3$  dB improvement in SNR — a significant advantage for the low-noise neural recording application.